

Supplementary Table: Full SPARC fitting results

Galaxy	N	R_L^2	$\chi_{\nu,L}^2$	AIC _L	R_M^2	$\chi_{\nu,M}^2$	AIC _M	R_N^2	$\chi_{\nu,N}^2$	AIC _N	k_C	p_{LRT}
CamB	9	0.974	0.593	13.8	−1.000	37.148	264.0	0.768	3.190	23.9	0.50	0.383
D564-8	6	0.999	0.047	12.0	0.483	6.253	29.0	0.962	0.914	9.8	0.47	0.959
D631-7	16	0.951	2.619	38.2	0.726	15.440	220.2	0.934	4.109	57.3	0.00	1.000
DDO064	14	0.983	0.282	14.3	0.954	0.348	8.2	0.942	0.665	14.6	0.84	0.279
DDO154	12	0.997	0.168	13.0	0.963	1.310	17.1	0.965	1.566	20.5	6.45	0.440
DDO161	31	0.995	0.223	17.6	0.254	47.969	1395.1	0.991	0.398	18.8	0.58	0.859
DDO168	10	0.973	2.581	22.3	0.803	10.453	87.6	0.906	6.404	46.4	7.53	1.000
DDO170	8	0.985	1.310	14.6	−0.030	40.878	249.3	0.957	1.589	14.4	0.00	0.997
ESO079-G014	15	0.995	0.305	14.7	0.961	3.475	49.2	0.961	4.161	53.8	0.00	1.000
ESO116-G012	15	0.996	0.479	16.3	0.954	4.952	68.4	0.975	2.727	38.0	0.00	1.000
ESO444-G084	7	0.993	0.677	12.7	0.834	6.900	38.5	0.973	0.830	10.5	0.46	0.867
ESO563-G021	30	0.967	5.913	153.9	0.924	17.368	490.3	0.928	18.521	489.5	0.00	1.000
F561-1	6	0.977	0.167	12.2	−1.000	19.959	83.8	0.860	0.901	9.8	0.00	1.000
F563-1	17	0.940	0.539	17.9	0.872	1.090	20.4	0.874	1.232	24.0	0.00	1.000
F563-V1	6	0.999	0.005	12.0	−1.000	30.658	126.6	0.916	0.386	8.8	2.87	0.971
F563-V2	10	0.991	0.024	12.1	0.969	0.398	7.2	0.985	0.457	10.7	0.65	0.954
F565-V2	7	0.991	0.410	12.4	0.935	0.441	6.2	0.948	0.619	9.9	0.00	0.999
F567-2	5	0.943	0.427	12.4	−1.000	5.051	19.2	0.849	1.063	9.1	0.00	0.999
F568-1	12	0.982	0.241	13.4	0.937	0.830	12.3	0.939	1.007	16.1	0.00	1.000
F568-3	18	0.964	0.664	20.0	0.919	2.729	47.7	0.786	5.012	78.2	0.00	1.000
F568-V1	15	0.984	0.105	12.9	0.914	0.317	8.1	0.945	0.162	9.8	0.00	1.000
F571-8	13	0.983	1.678	23.7	0.499	47.881	530.7	0.963	3.688	41.2	1.29	1.000
F571-V1	7	0.999	0.048	12.0	0.964	0.319	5.6	0.942	0.531	9.6	0.00	1.000
F574-1	14	0.996	0.185	13.5	0.925	2.298	31.6	0.913	3.743	45.4	6.58	0.875
F574-2	5	0.994	0.035	12.0	−1.000	19.079	61.2	0.970	0.147	8.1	9.99	0.906
F579-V1	14	0.994	0.041	12.3	0.444	2.692	36.3	0.962	0.225	10.3	0.00	1.000
F583-1	25	0.989	0.207	15.9	0.881	2.068	51.6	0.922	1.676	43.2	1.90	0.582
F583-4	12	0.989	0.300	13.8	0.993	0.127	5.3	0.928	1.172	17.4	0.00	1.000
IC2574	34	0.981	0.642	30.0	0.891	3.632	120.2	0.748	8.015	248.5	5.69*	0.000
IC4202	32	0.973	3.376	99.8	0.798	30.465	918.0	0.881	19.432	552.1	0.00*	0.000
KK98-251	15	0.996	0.173	13.6	0.189	23.628	311.2	0.947	1.831	28.1	0.00	1.000
NGC0024	29	0.989	0.230	17.3	0.967	0.691	22.6	0.973	0.359	17.0	0.00	1.000
NGC0055	21	0.997	0.117	13.8	0.825	6.013	118.3	0.681	13.289	233.9	10.61	0.418
NGC0100	21	0.996	0.093	13.4	0.941	1.408	30.8	0.481	11.317	200.4	0.00	1.000
NGC0247	26	0.995	0.299	18.0	0.929	2.440	62.5	0.799	2.403	60.9	24.20*	0.000
NGC0289	28	0.099	2.601	69.2	−1.000	8.188	216.9	−0.098	1.996	55.9	0.00	1.000
NGC0300	25	0.966	0.480	21.1	0.938	1.250	32.7	−0.090	29.141	620.0	0.00	1.000
NGC0801	13	0.823	9.690	79.8	0.061	41.321	458.5	0.856	6.041	62.4	0.00	1.000

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Galaxy	N	R_L^2	$\chi_{\nu,L}^2$	AIC _L	R_M^2	$\chi_{\nu,M}^2$	AIC _M	R_N^2	$\chi_{\nu,N}^2$	AIC _N	k_C	p_{LRT}
NGC0891	18	-0.407	2.596	43.2	-1.000	7.739	127.8	-0.124	2.104	37.5	8.27	0.409
NGC1003	36	0.935	2.983	101.5	0.886	3.995	139.8	0.930	2.091	74.9	13.93	0.161
NGC1090	24	0.987	1.039	30.7	0.881	7.930	178.5	0.942	2.639	60.8	1.64	0.875
NGC1705	14	0.985	0.028	12.2	0.691	0.308	7.7	0.917	0.078	8.8	6.26	0.741
NGC2366	26	0.991	0.330	18.6	0.679	7.421	182.1	0.923	2.391	60.6	0.54	0.403
NGC2403	73	0.987	1.599	119.1	0.957	4.733	340.0	0.983	2.190	159.1	0.00	1.000
NGC2683	11	0.832	1.558	19.8	0.575	2.557	27.0	0.875	0.936	14.6	0.00	1.000
NGC2841	50	0.422	4.538	211.7	0.764	1.894	94.9	0.867	1.381	71.5	69.55*	0.003
NGC2903	34	0.990	0.465	25.0	0.624	11.575	374.4	0.778	6.065	189.9	40.26*	0.002
NGC2915	30	0.943	0.259	18.2	0.015	3.083	90.3	0.815	0.720	26.7	0.00	1.000
NGC2955	24	0.849	2.095	49.7	0.688	3.607	83.4	0.770	3.178	71.6	0.00	1.000
NGC2976	27	0.996	0.295	18.2	0.976	1.299	36.5	0.981	0.875	28.1	0.00	1.000
NGC2998	13	0.987	0.641	16.5	0.348	8.054	92.6	0.900	1.638	22.7	6.35	0.592
NGC3109	25	0.999	0.069	13.3	0.943	3.699	89.1	0.958	3.648	84.6	0.16	0.896
NGC3198	43	0.991	0.680	37.2	0.927	9.683	401.0	0.936	1.348	60.6	7.78	0.269
NGC3521	41	0.955	0.074	14.6	0.809	0.540	25.1	0.916	0.246	17.1	26.23	0.278
NGC3726	12	0.847	4.033	36.2	0.780	3.563	39.6	0.859	2.457	27.7	0.00	0.772
NGC3741	21	0.992	0.283	16.2	0.950	1.258	27.9	0.987	0.383	14.5	0.95	0.480
NGC3769	12	0.347	2.073	24.4	0.579	0.834	12.3	0.856	0.533	12.3	0.00	1.000
NGC3877	13	0.986	0.590	16.1	0.787	8.021	92.2	0.868	4.376	47.4	0.00	1.000
NGC3893	10	0.962	0.371	13.5	0.772	1.008	12.1	0.869	0.620	11.7	33.13	0.744
NGC3917	17	0.995	0.523	17.8	0.960	3.208	52.1	0.966	3.066	47.9	0.00	1.000
NGC3949	7	0.976	0.299	12.3	0.819	0.418	6.1	0.976	0.193	8.6	0.00	1.000
NGC3953	8	0.993	0.132	12.3	0.439	3.288	23.7	0.946	0.576	10.3	59.25	0.884
NGC3972	10	0.992	0.720	14.9	0.963	1.568	16.5	0.975	1.387	16.3	23.35	0.329
NGC3992	9	0.860	1.382	16.1	0.120	4.110	32.8	0.909	0.574	10.9	80.00	0.901
NGC4010	12	0.991	0.560	15.4	0.930	2.323	27.2	0.947	2.605	28.8	19.26	0.233
NGC4013	36	0.523	2.472	86.2	0.604	1.494	54.8	0.785	0.724	31.2	80.00*	0.000
NGC4051	7	0.922	2.194	14.2	0.659	2.425	16.1	0.914	1.073	11.2	80.00	0.386
NGC4068	6	0.997	0.203	12.2	0.865	5.042	24.2	0.929	2.913	13.8	0.85	0.690
NGC4085	7	0.994	1.387	13.4	0.849	5.030	29.2	0.910	5.365	24.1	0.00	1.000
NGC4088	12	0.966	0.893	17.4	0.918	1.172	15.7	0.962	0.523	12.2	0.00	1.000
NGC4100	24	0.964	0.290	17.2	0.733	2.012	48.3	0.875	0.943	26.9	0.00	0.662
NGC4138	7	0.799	1.599	13.6	0.681	1.240	10.2	0.786	1.021	11.1	0.00	1.000
NGC4157	17	0.923	0.741	20.2	0.935	0.642	13.6	0.970	0.245	11.2	0.00	1.000
NGC4183	23	0.988	0.117	14.0	0.736	2.810	63.0	0.354	5.569	113.8	16.68	0.522
NGC4214	14	0.953	0.140	13.1	0.145	1.382	20.6	0.639	0.661	14.6	9.23	1.000
NGC4217	19	0.993	0.223	14.9	0.935	1.610	31.4	0.933	1.845	35.7	6.11	0.821
NGC4389	6	0.994	0.514	12.5	0.824	4.534	22.1	0.658	16.926	41.9	10.96	0.612
NGC4559	32	0.984	0.211	17.5	0.844	1.672	54.1	0.981	0.201	13.6	0.00	0.577
NGC5005	18	0.907	0.112	13.3	0.823	0.055	4.9	0.837	0.057	8.8	0.00	1.000

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Galaxy	N	R_L^2	$\chi_{\nu,L}^2$	AIC _L	R_M^2	$\chi_{\nu,M}^2$	AIC _M	R_N^2	$\chi_{\nu,N}^2$	AIC _N	k_C	p_{LRT}
NGC5033	22	0.783	2.767	56.3	0.133	8.657	177.1	0.859	2.721	57.0	0.00	1.000
NGC5055	28	0.829	3.590	91.0	0.704	10.446	275.6	0.980	0.930	30.3	63.81*	0.000
NGC5371	19	0.262	9.122	130.6	-1.000	58.036	990.6	-0.164	4.533	76.0	0.00	1.000
NGC5585	24	0.995	0.590	22.6	0.961	4.609	105.4	0.324	96.823	1944.5	0.45	0.157
NGC5907	19	0.705	3.759	60.9	-1.000	28.476	488.1	0.504	4.272	72.1	63.33	0.721
NGC5985	33	0.875	1.339	48.2	0.326	6.226	197.0	0.845	1.748	58.7	0.00	1.000
NGC6015	44	0.962	3.360	139.7	0.880	12.204	516.6	0.918	7.328	301.1	0.00	1.000
NGC6195	23	0.666	1.698	40.9	0.704	1.839	42.6	0.684	1.779	41.8	1.04	0.955
NGC6503	31	0.967	0.351	20.8	0.746	2.230	68.7	0.844	1.164	39.4	62.51*	0.011
NGC6674	15	-0.343	15.936	155.4	-1.000	8.986	120.8	-0.576	2.384	34.2	80.00	1.000
NGC6946	58	0.902	1.521	91.1	0.807	1.854	107.8	0.895	1.510	89.5	30.87*	0.000
NGC7331	36	0.342	1.022	42.7	0.593	1.307	48.4	0.784	0.689	30.0	80.00	0.108
NGC7793	46	0.986	0.289	23.5	0.961	0.850	41.4	0.953	0.920	46.6	0.00	1.000
NGC7814	18	0.971	0.463	17.6	0.754	1.314	25.0	0.949	0.410	13.7	49.08*	0.000
PGC51017	6	0.992	0.008	12.0	-1.000	101.493	410.0	-1.000	1.772	11.5	3.29	0.604
UGC00128	22	0.988	1.422	34.8	0.733	30.820	620.4	0.961	1.407	33.3	0.00	1.000
UGC00191	9	0.989	1.067	15.2	0.853	8.784	65.5	0.723	41.963	217.8	25.70	0.053
UGC00731	12	0.993	0.274	13.6	0.418	14.626	150.3	0.998	0.087	8.7	0.62	0.886
UGC00891	5	0.997	0.706	12.7	0.937	3.837	15.5	0.967	4.262	12.3	0.00	1.000
UGC01230	11	0.966	0.494	14.5	0.478	4.486	44.4	0.641	3.881	35.2	0.00	1.000
UGC01281	25	0.997	0.069	13.3	0.988	0.230	9.3	0.940	1.168	32.5	0.36	0.441
UGC02023	5	0.998	0.032	12.0	0.927	0.373	5.1	0.753	3.779	11.8	0.60	1.000
UGC02259	8	0.991	0.701	13.4	0.694	5.718	38.3	0.982	0.695	10.8	0.00	0.994
UGC02455	8	0.931	4.062	20.1	-0.492	29.166	179.0	0.587	12.122	56.5	4.77	1.000
UGC02487	17	0.643	7.531	94.8	0.842	7.282	113.2	0.717	3.995	59.9	80.00	0.472
UGC02885	19	0.212	2.040	38.5	0.617	0.758	16.9	0.628	0.834	20.5	11.10	0.547
UGC02916	43	0.601	0.640	35.7	-1.000	8.870	367.7	-1.000	7.576	303.5	0.00	1.000
UGC02953	115	-0.917	10.219	1125.8	0.398	3.921	447.0	0.395	3.808	430.7	80.00*	0.000
UGC03205	48	0.919	1.868	90.4	0.678	4.057	190.6	0.770	3.267	151.8	15.69*	0.046
UGC03546	30	0.887	0.587	26.1	0.931	0.537	19.0	0.923	0.450	19.7	71.90	0.071
UGC03580	47	-0.443	2.427	111.5	0.780	2.197	102.9	0.763	1.545	74.4	36.73*	0.000
UGC04278	25	0.986	0.544	22.3	0.930	2.522	62.0	0.736	8.121	178.6	0.00	1.000
UGC04305	22	0.940	0.494	19.9	-1.000	42.771	859.4	0.747	1.882	41.9	0.55	0.811
UGC04325	8	0.999	0.129	12.3	0.804	7.830	51.0	0.959	2.451	17.8	3.67	0.904
UGC04483	8	0.989	0.449	12.9	0.680	4.284	29.7	0.967	0.583	10.3	0.18	0.682
UGC04499	9	0.997	0.292	12.9	0.904	3.573	29.0	0.963	1.128	13.6	0.00	1.000
UGC05005	11	0.998	0.028	12.1	0.936	0.604	9.4	0.978	0.247	9.7	7.08	0.603
UGC05253	73	0.678	0.464	43.1	-1.000	3.876	279.2	-1.000	2.082	151.6	36.73*	0.000
UGC05414	6	1.000	0.093	12.1	0.998	0.081	4.3	0.838	15.687	39.4	11.81	0.675
UGC05716	12	0.993	0.418	14.5	0.750	8.478	88.8	0.989	0.411	11.3	1.84	0.860
UGC05721	23	0.979	0.331	17.6	0.789	2.787	62.5	0.934	0.926	25.6	0.00	1.000

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Galaxy	N	R_L^2	$\chi_{\nu,L}^2$	AIC _L	R_M^2	$\chi_{\nu,M}^2$	AIC _M	R_N^2	$\chi_{\nu,N}^2$	AIC _N	k_C	$PLRT$
UGC05750	11	0.995	0.103	12.5	0.843	1.171	14.5	0.881	1.212	16.5	0.00	1.000
UGC05764	10	0.989	0.583	14.3	0.823	4.109	36.9	0.900	3.662	30.0	6.27*	0.021
UGC05829	11	0.987	0.296	13.5	0.758	2.954	30.6	0.994	0.097	8.7	12.60	0.293
UGC05918	8	0.992	0.158	12.3	0.791	1.347	12.1	0.986	0.131	8.5	0.00	0.999
UGC05986	15	0.998	0.153	13.4	0.943	6.332	86.3	0.923	5.838	72.2	7.06	0.618
UGC05999	5	0.988	0.955	13.0	0.808	4.932	18.8	0.916	6.474	14.5	0.00	0.794
UGC06399	9	0.998	0.097	12.3	0.976	0.376	6.6	0.858	3.816	27.1	2.57	1.000
UGC06446	17	0.979	0.209	14.3	0.843	0.952	18.3	0.968	0.230	11.0	0.00	1.000
UGC06614	13	0.258	1.748	24.2	0.508	0.711	11.8	0.893	0.141	9.3	0.00	1.000
UGC06628	7	0.998	0.006	12.0	-1.000	8.617	47.1	0.572	0.392	9.2	0.00	1.000
UGC06667	9	0.993	0.769	14.3	0.935	3.122	25.9	0.974	1.397	15.0	0.90	0.877
UGC06786	45	0.973	0.181	19.1	0.034	9.591	416.4	0.500	0.470	27.3	21.34	0.072
UGC06787	71	0.505	10.361	685.5	-0.062	16.695	1155.9	0.492	7.855	534.3	13.72*	0.002
UGC06818	8	0.979	1.161	14.3	0.834	3.064	22.4	0.905	2.927	19.7	0.00	1.000
UGC06917	11	0.989	0.327	13.6	0.941	1.132	14.2	0.968	0.799	13.6	0.00	1.000
UGC06923	6	0.983	0.813	12.8	0.908	0.957	7.8	0.305	12.207	32.4	0.00	1.000
UGC06930	10	0.990	0.123	12.5	0.688	2.219	21.8	0.441	4.125	32.8	0.00	1.000
UGC06973	9	0.625	0.510	13.5	-1.000	15.461	112.2	0.559	0.418	10.1	2.88	0.894
UGC06983	17	0.959	0.272	15.0	0.680	1.251	22.8	0.883	0.665	16.6	0.00	1.000
UGC07089	12	0.994	0.139	12.8	0.990	0.161	5.6	0.738	3.701	37.6	14.05	0.401
UGC07125	13	0.994	0.213	13.5	-1.000	86.329	953.6	0.956	1.011	17.1	0.00	1.000
UGC07151	11	0.989	0.869	16.3	0.900	3.902	39.1	0.871	6.251	51.8	1.60	0.840
UGC07261	7	0.998	0.034	12.0	0.758	0.927	8.6	0.914	0.551	9.7	0.00	1.000
UGC07323	10	0.998	0.155	12.6	0.991	0.329	6.6	0.931	3.201	27.2	0.00	1.000
UGC07399	10	0.998	0.168	12.7	0.859	4.939	43.5	0.981	1.172	15.0	33.05	0.248
UGC07524	31	0.996	0.191	16.8	0.897	3.765	113.2	0.928	2.938	87.3	0.00	1.000
UGC07559	7	0.998	0.092	12.1	0.461	4.553	26.8	0.961	0.543	9.6	3.45	0.554
UGC07577	9	0.978	0.133	12.4	-1.000	10.360	76.5	0.959	0.149	8.7	0.65	0.308
UGC07603	12	0.998	0.077	12.5	0.872	3.388	37.9	0.946	1.779	22.2	2.91	0.577
UGC07608	8	0.996	0.111	12.2	0.933	0.601	7.6	0.953	0.636	10.5	8.90	0.898
UGC07690	7	0.972	0.247	12.2	-0.309	2.279	15.4	0.810	0.551	9.7	10.88	0.375
UGC07866	7	0.995	0.085	12.1	0.641	1.319	10.6	0.991	0.055	8.2	0.00	1.000
UGC08286	17	0.991	0.713	19.8	0.953	2.563	42.5	0.968	1.789	31.3	0.00	1.000
UGC08490	30	0.993	0.055	13.3	0.962	0.252	11.1	0.983	0.121	11.1	6.66	0.117
UGC08550	11	0.986	0.529	14.6	0.929	1.651	18.9	0.976	0.667	12.7	0.00	1.000
UGC08699	41	0.658	0.913	43.9	-0.296	0.867	37.8	0.246	0.628	31.2	23.38*	0.001
UGC08837	8	0.995	0.578	13.2	0.754	11.082	70.5	0.824	9.438	45.8	0.00	1.000
UGC09037	22	0.948	0.569	21.1	0.859	1.636	36.7	-0.546	28.489	520.8	0.00	0.997
UGC09133	68	0.795	7.972	506.2	0.177	19.598	1297.5	0.333	3.676	243.2	80.00*	0.000
UGC09992	5	0.969	0.023	12.0	-1.000	5.237	19.7	0.958	0.032	8.0	0.00	1.000
UGC10310	7	0.996	0.184	12.2	0.702	2.930	18.6	0.935	1.072	11.2	2.68	0.958

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Galaxy	N	R_L^2	$\chi_{\nu,L}^2$	AIC _L	R_M^2	$\chi_{\nu,M}^2$	AIC _M	R_N^2	$\chi_{\nu,N}^2$	AIC _N	k_C	p_{LRT}
UGC11455	36	0.984	1.806	66.2	0.966	4.772	166.2	0.964	5.259	176.3	1.06	0.791
UGC11557	12	0.987	0.321	13.9	0.891	1.609	20.1	0.845	2.564	28.5	0.00	1.000
UGC11820	10	0.964	2.388	21.6	0.839	18.263	150.1	0.989	0.423	10.5	0.00	1.000
UGC11914	65	0.972	0.549	44.4	0.832	2.786	179.5	0.837	0.664	48.5	7.29	0.443
UGC12506	31	0.965	0.194	16.8	0.314	2.705	82.5	0.971	0.157	12.2	20.51	0.497
UGC12632	15	0.997	0.071	12.6	0.740	7.304	98.9	0.980	0.277	11.0	0.00	1.000
UGC12732	16	0.997	0.126	13.3	0.842	4.236	63.3	0.996	0.129	9.5	15.65*	0.008
UGCA281	7	0.999	0.109	12.1	0.995	0.112	4.6	0.991	0.295	8.9	0.40	0.288
UGCA442	8	0.999	0.065	12.1	0.965	2.254	17.5	0.978	2.286	17.1	0.18	0.992
UGCA444	36	0.988	0.057	13.7	0.724	1.956	70.5	0.988	0.063	10.0	0.51	0.213